Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B.TECH (ECE)-VII
Name of the Paper: High Speed Communication Circuits
Paper Code: TEC-703
Time: 1.5 hour

Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.

Q1. **(10 Marks) CO 1**

a. Discuss the types of noise in communication systems. Differentiate between internal and external noise, and explain how each affects the performance of amplifiers.

OR

b. Illustrate and describe the ideal amplifier model. Discuss its key components, operational principles, and explain its gain characteristics.

Q2. **(10 Marks) CO 1**

a. Discuss the different types of coupling methods used in amplifier circuits. Explain each type in detail, highlighting their working principles, advantages and disadvantages.

OR

b. Discuss power amplifiers in detail, including their classifications, operating principles, and typical applications.

Q3. **(10 Marks) CO 1,2**

a.Provide brief notes on different LAN topologies. Discuss their structure, advantages, disadvantages, and common use cases."

OR

b. Compare and contrast mixing and adding in the context of signal processing. Highlight their differences in terms of functionality, applications, and effects on signals."

Q4. **(10 Marks) CO 2**

a. What are cascaded amplifiers? Explain their working principle in detail, including their design.

OR

b. Describe the construction of a two-stage RC-coupled amplifier. Explain its working principle, key components, and discuss its advantages and typical applications.

Q5. **(10 Marks) CO 2**

a. What is negative feedback in an amplifier. Briefly explain the types of negative feedback circuit.

OR

b. List down the various advantages of RC Coupled Amplifier.